



GWT-2.39-256144-AWMN-04

Ver:1.1

MONO TFT Display Module

< ◇ > Preliminary Specification

< ◆ > Finally Specification

CUSTOMER'S APPROVAL	
CUSTOMER :	
SIGNATURE:	DATE:

APPROVED BY	PM REVIEWED	PD REVIEWED	PREPARED BY

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Never stop production, stock for sale, support small quantity, full of wide products range

Revision History

[illegible]

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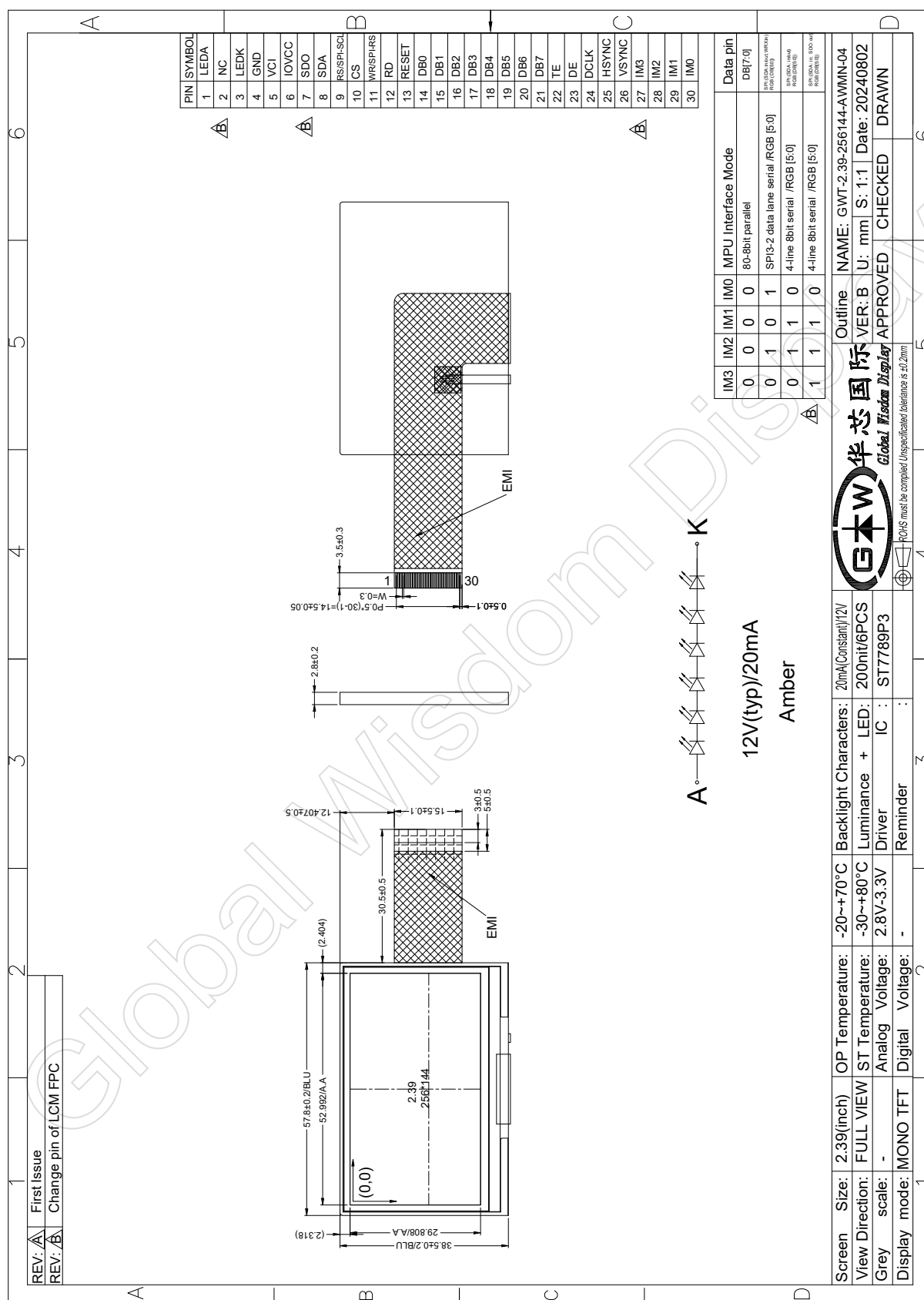
1. General Description

Features	Details	Unit
Display Size(Diagonal)	2.39"	
Display type	64 Gray scale TFT(MONO TFT)	
Display Mode	Normally Black	
Number of dots	256 x 144	
View Direction	ALL	
Module Outline	57.8(H) x 38.5(V) x 2.8(T) (Note1)	mm
Active Area	52.992(H) x29.808(V)	mm
Dot Pitch	207(H) x 207(V)	um
Polarizer Surface Treatment	Anti-glare	
LCM Driver IC	ST7789P3	
LCM Interface	SPI+RGB/MCU interface	
Operating Temperature	-20~70	°C
Storage Temperature	-30~80	°C
Weight	-	g

Note 1: Inclusive hooks, posts, FFC/FPC tail etc.



1.1. Outline Drawing



2. Absolute Maximum Ratings

The followings are maximum values which, if exceed, may cause faulty operation or damage to the unit.
The operational and non-operational maximum voltage and current values are listed in Table.

 $V_{SS}=0V$, $T_a=25^{\circ}C$

Item	Symbol	Min.	Max.	Unit
Supply Voltage	VCI	-0.3	+4.6	V
Power Supply for I/O System.	IOVCC	-0.3	+4.0	
Operating temperature	T_{OP}	-20	70	$^{\circ}C$
Storage temperature	T_{STG}	-30	80	$^{\circ}C$

Note 1: If T_a below $50^{\circ}C$, the maximal humidity is 90%RH, if T_a over $50^{\circ}C$, absolute humidity should be less than 60%RH.

Note 2: The response time will be extremely slow when the operating temperature is around $-10^{\circ}C$, and the back ground will become darker at high temperature operating.

Note 3: These range above is maximum value not the actual operating temperature . Actual Operating temperature is no more than $40^{\circ}C$ and temperature refers to the LCM surface temperature ;

Note 4: GWD is not responsible for product problems beyond the use conditions.

3. DC Characteristics

Item		Symbol	Min.	Typ.	Max.	Unit
Supply Voltage		V _{CI}	2.4	2.75	3.3	V
Power Supply for I/O System.		IOVCC	1.65	1.8	3.3	
Logic Low input voltage		V _{IL}	GND	-	0.3*IOVCC	V
Logic High input voltage		V _{IH}	0.7*IOVCC	-	IOVCC	V
Logic Low output voltage		V _{OL}	GND	-	0.2*IOVCC	
Logic High output voltage		V _{OH}	0.8*IOVCC	-	IOVCC	
Current Consumption All Black	Logic	I _{CC} + I _{IN}	-	TBD	-	mA
	Analog					

Note 1: All of the voltage listed above are with respective to GND = 0v

Note 2: Device is subject to be damaged permanently if stresses beyond those absolute maximum rating listed above

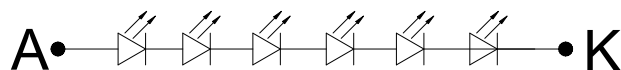
4. Backlight Characteristic

4.1. Backlight Characteristic

Item	Symbol	Condition	Min.	Typ.	Max.	Unit
Forward Voltage	V_F	$T_a=25^{\circ}C$, $I_F=20mA/LED$	-	12	-	V
Forward Current	I_F	$T_a=25^{\circ}C$, $V_F=2V/LED$	-	20	-	mA
Power dissipation	P_D	-	-	240	-	mW
Uniformity	Avg	-	-	80	-	%
LED working life($25^{\circ}C$)	-	-	-	30,000	-	Hrs
Drive method	Constant current					
LED Configuration	6 Amber LEDs in one string					

Note1: LED life time defined as follows: The final brightness is at 50% of original brightness.
The environmental conducted under ambient air flow, at $T_a=25 \pm 2^\circ\text{C}$, $60\%RH \pm 5\%$, $I_F=20\text{mA}$.

4.2. Backlighting circuit



5. Optical Characteristics

 $T_a=25^\circ\text{C}$, $V_{DD}=3.3\text{V}$

Item		Symbol	Condition	Specification			Unit	Note
				Min.	Typ.	Max.		
Luminance on TFT(I_f =20mA/LED)		Lv		150	200	-	cd/m²	(1)
Contrast ratio		CR		-	1000	-		(2)
Luminance uniformity		Lu		-	80	-	%	(3)
Response time		TR+TF		-	25	-	ms	(4)
LED Dominant Wavelength				600	-	612	nm	
Viewing Angle	Horizontal	θx+	Center CR≥10	-	80	-	Deg.	(5)
		θx-		-	80	-		
	Vertical	φY+		-	80	-		
		φY-		-	80	-		

Note:

(1) Luminance measurement

The data are measured at the center point of LCM after BL are lighted on and LCM displays are fully white

(2) The definition of Contrast Ratio (CR)

$$\text{Contrast Ratio (CR)} = \frac{\text{Luminance When LCD is at "White" state}}{\text{Luminance When LCD is at "Black" state}}$$

(3) Luminance uniformity

The luminance uniformity is calculated by using following formula:

$$Lu = Lv(\text{Min}) / Lv(\text{Max}) \times 100\%$$

Where: $Lv(\text{Max})$ = Maximum brightness in 9 measured spots

$Lv(\text{Min})$ = Minimum brightness in 9 measured spots.

9 measurement points are illustrated below:

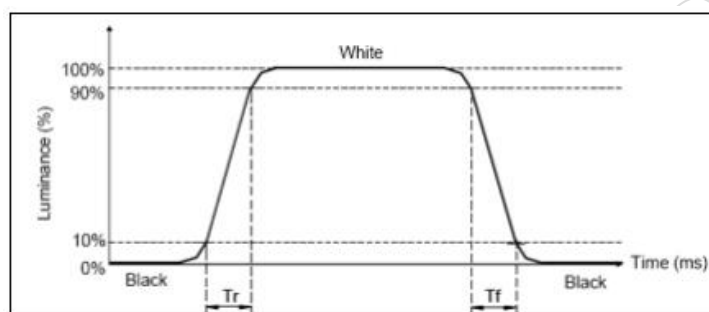
L: Length of Active Area

W:Width of Active Area



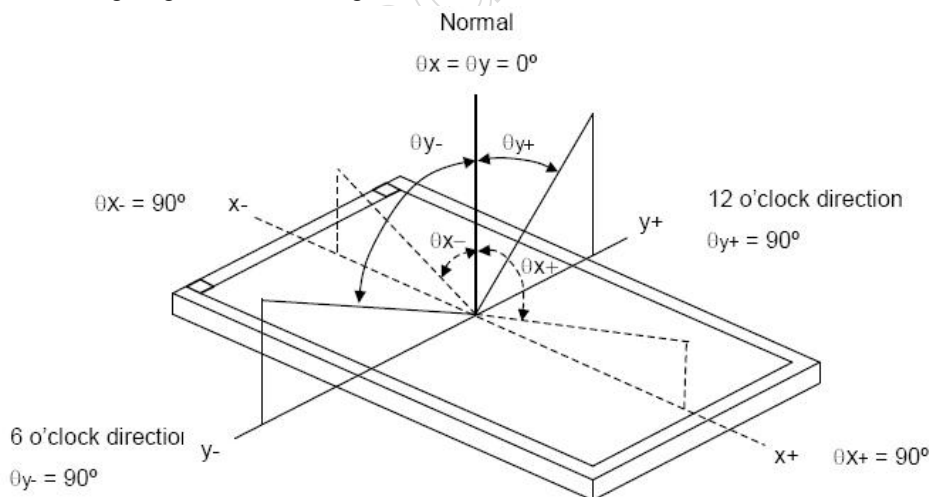
(4) Response time

The response time is defined as the time interval between the 10% and 90% of amplitudes. Refer to figure as below.

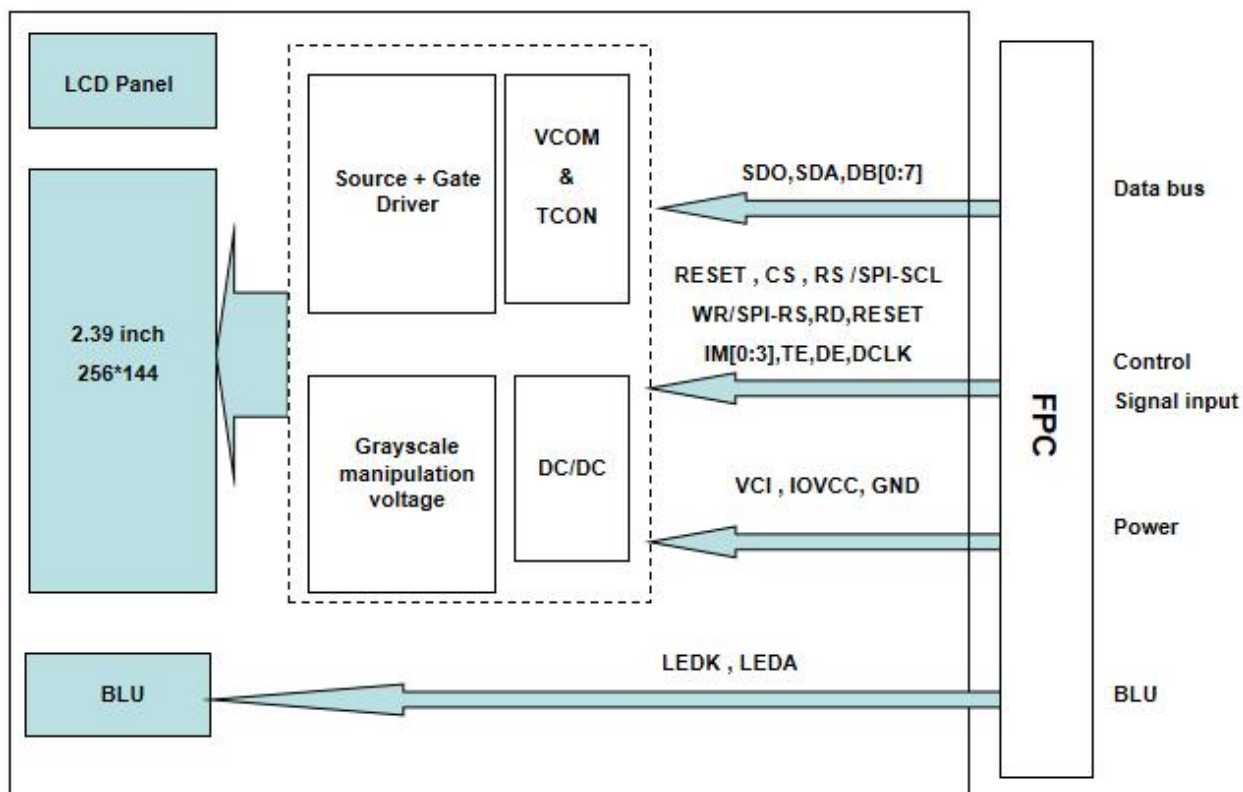


(5) Viewing angle

Definition of viewing angle θ , Refer to figure as below.



6. Block Diagram and Power Supply



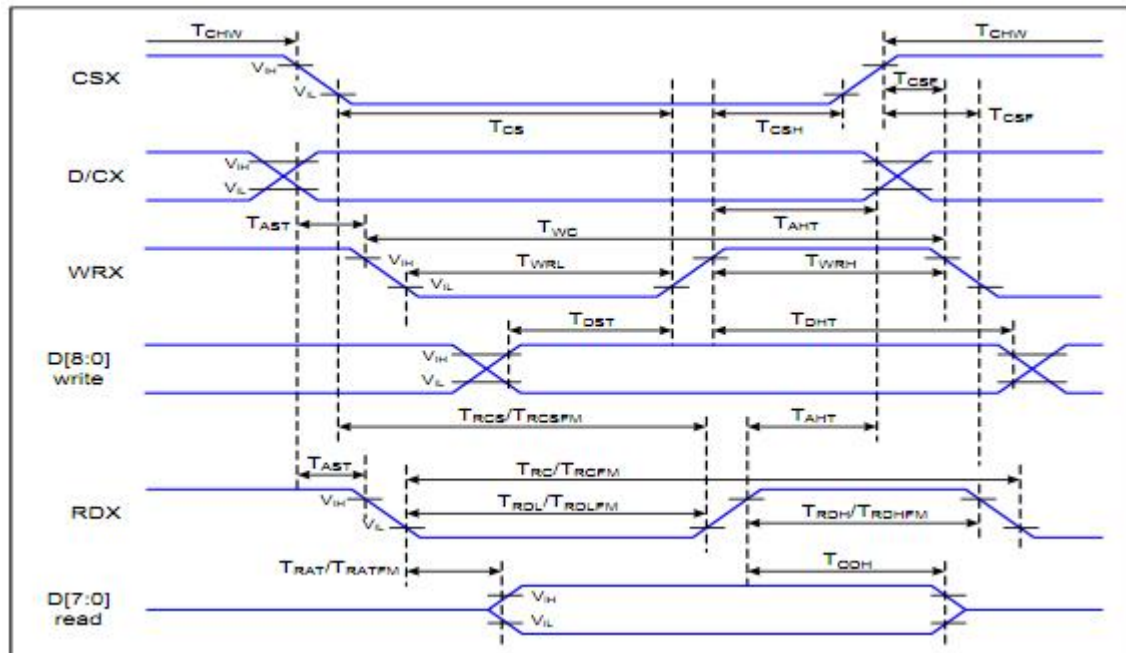


7. Interface Pins Definition

No.	Symbol	Function																														
1	LEDA	Power for LED Anode																														
2	NC	NC																														
3	LEDK	Power for LED Cathode																														
4	GND	Ground																														
5	VCI	Power Supply for Analog, Digital System and Booster Circuit																														
6	IOVCC	Power Supply for I/O System.																														
7	SDO	SPI interface output pin.																														
8	SDA	Serial data input/output pin. -When IM3: Low, SPI interface input/output pin. -When IM3: High, SPI interface input pin.																														
9	RS/SPI-SCL	Serial clock input pin.																														
10	CS	Chip select input pin.																														
11	WR/SPI-RS	Write enable in MCU parallel interface/Serial clock input pin.																														
12	RD	Read enable in 8080 MCU parallel interface																														
13	RESET	Reset input pin.																														
14-21	DB0-DB7	DB [7:0] are used as MCU parallel interface data bus DB [5:0] are used as RGB interface data bus																														
22	TE	Tearing effect signal is used to synchronize MCU to frame memory writing.																														
23	DE	Data enable signal for RGB interface operation.																														
24	DCLK	Dot clock signal for RGB interface operation																														
25	HSYNC	Horizontal (Line) synchronizing input signal for RGB interface operation.																														
26	VSYNC	Vertical (Frame) synchronizing input signal for RGB interface operation.																														
27	IM3	<table><tr><th>IM3</th><th>IM2</th><th>IM1</th><th>IM0</th><th>MPU Interface Mode</th><th>Data pin</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>80-8bit parallel</td><td>DB[7:0]</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>SPI3-2 data lane serial /RGB [5:0]</td><td>SPI(SDA:in/out, WRX:in, RGB:(DB[5:0])</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>4-line 8bit serial /RGB [5:0]</td><td>SPI(SDA: in/out, RGB:(DB[5:0])</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>4-line 8bit serial /RGB [5:0]</td><td>SPI(SDA: in, SDO: out, RGB:(DB[5:0])</td></tr></table>	IM3	IM2	IM1	IM0	MPU Interface Mode	Data pin	0	0	0	0	80-8bit parallel	DB[7:0]	0	1	0	1	SPI3-2 data lane serial /RGB [5:0]	SPI(SDA:in/out, WRX:in, RGB:(DB[5:0])	0	1	1	0	4-line 8bit serial /RGB [5:0]	SPI(SDA: in/out, RGB:(DB[5:0])	1	1	1	0	4-line 8bit serial /RGB [5:0]	SPI(SDA: in, SDO: out, RGB:(DB[5:0])
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28	IM2																															
29	IM1																															
30	IM0																															

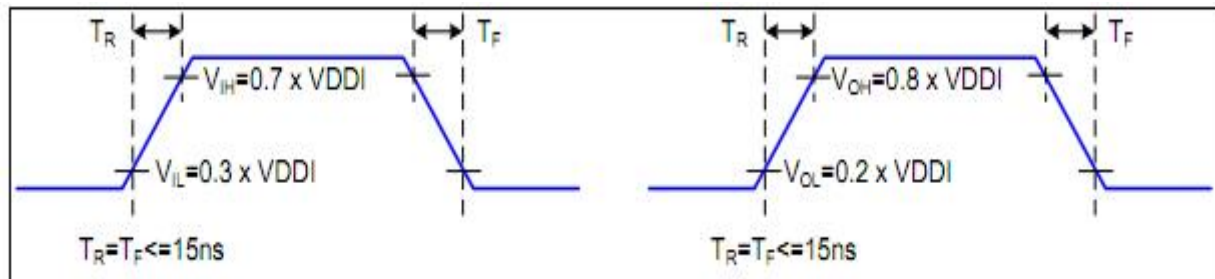
8. AC Characteristics

8.1. 8080 Series MCU Parallel Interface Characteristics: 9/8-bit Bus

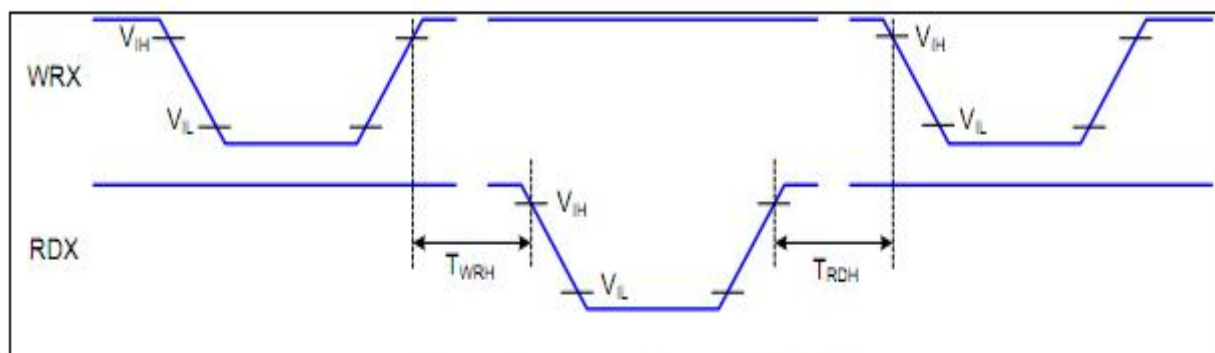


Signal	Symbol	Parameter	Min	Max	Unit	Description
D/CX	T_{AST}	Address setup time	0	-	ns	-
	T_{AHT}	Address hold time (Write/Read)	10	-	ns	
CSX	T_{CHW}	Chip select "H" pulse width	0	-	ns	-
	T_{CS}	Chip select setup time (Write)	15	-	ns	
	T_{RCS}	Chip select setup time (Read ID)	45	-	ns	
	T_{RCSFM}	Chip select setup time (Read FM)	355	-	ns	
	T_{CSF}	Chip select wait time (Write/Read)	10	-	ns	
	T_{CSH}	Chip select hold time	10	-	ns	
WRX	T_{WC}	Write cycle	66	-	ns	-
	T_{WRH}	Control pulse "H" duration	15	-	ns	
	T_{WRL}	Control pulse "L" duration	15	-	ns	
RDX (ID)	T_{RC}	Read cycle (ID)	160	-	ns	When read ID data
	T_{RDH}	Control pulse "H" duration (ID)	90	-	ns	
	T_{RDL}	Control pulse "L" duration (ID)	45	-	ns	
RDX (FM)	T_{RCFM}	Read cycle (FM)	450	-	ns	When read from frame memory
	T_{RDHF}	Control pulse "H" duration (FM)	90	-	ns	
	T_{RDLF}	Control pulse "L" duration (FM)	350	-	ns	
D[7:0]	T_{DST}	Data setup time	10	-	ns	For CL=30pF

T_{DHT}	Data hold time	10	-	ns
T_{RAT}	Read access time (ID)	-	40	ns
T_{RATFM}	Read access time (FM)	-	340	ns
T_{ODH}	Output disable time	20	80	ns

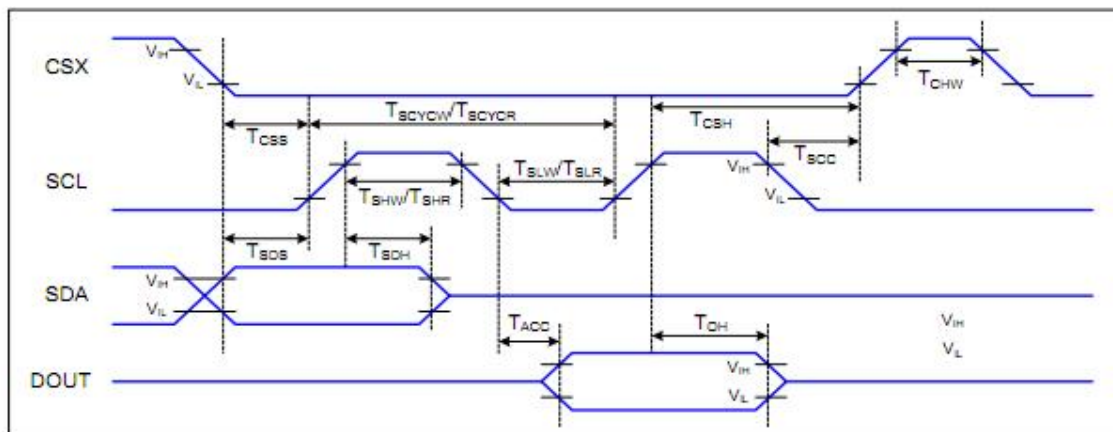


Rising and Falling Timing for I/O Signal



Write-to-Read and Read-to-Write Timing

8.2. Serial Interface Characteristics (3-line serial)



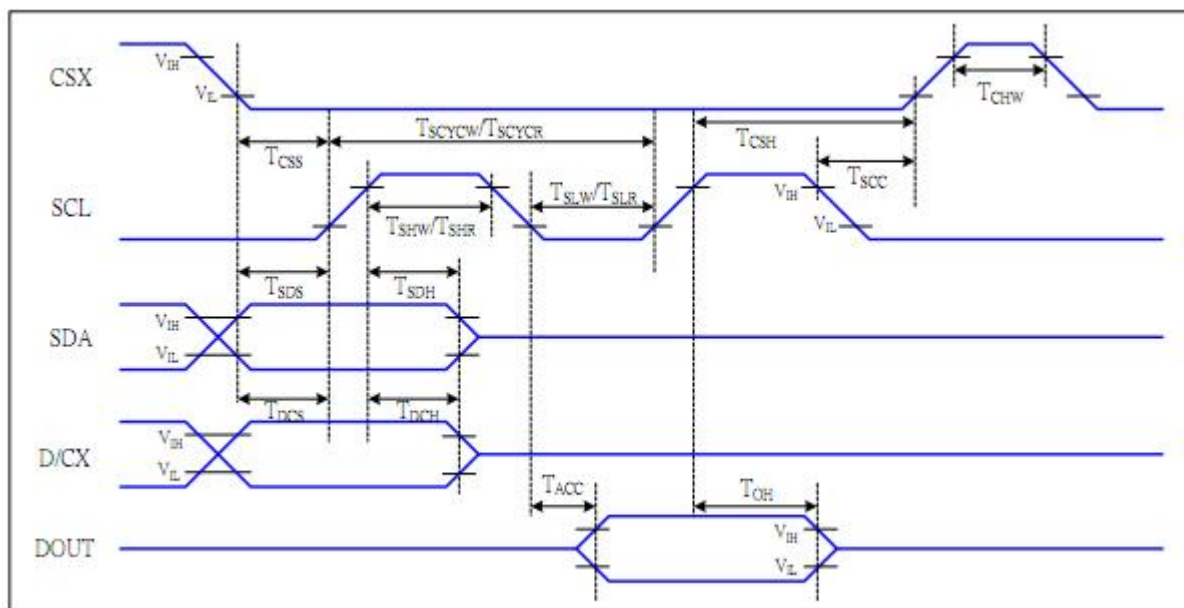
3 - line serial Interface Timing Characteristics

Signal	Symbol	Parameter	Min	Max	Unit	Description
CSX	T _{CSS}	Chip select setup time (write)	15	-	ns	-
	T _{CSH}	Chip select hold time (write)	15	-	ns	
	T _{CSS}	Chip select setup time (read)	60	-	ns	
	T _{SCC}	Chip select hold time (read)	65	-	ns	
	T _{CHW}	Chip select "H" pulse width	40	-	ns	
SCL	T _{SCYCW}	Serial clock cycle (Write)	16	-	ns	-
	T _{SHW}	SCL "H" pulse width (Write)	7	-	ns	
	T _{SLW}	SCL "L" pulse width (Write)	7	-	ns	
	T _{SCYCR}	Serial clock cycle (Read)	150	-	ns	
	T _{SHR}	SCL "H" pulse width (Read)	60	-	ns	
	T _{SLR}	SCL "L" pulse width (Read)	60	-	ns	
SDA (DIN)	T _{SDS}	Data setup time	7	-	ns	-
	T _{SDH}	Data hold time	7	-	ns	
DOUT	T _{ACC}	Access time	10	50	ns	For maximum CL=30pF
	T _{OH}	Output disable time	15	50	ns	For minimum CL=8pF

3-line serial Interface Characteristics

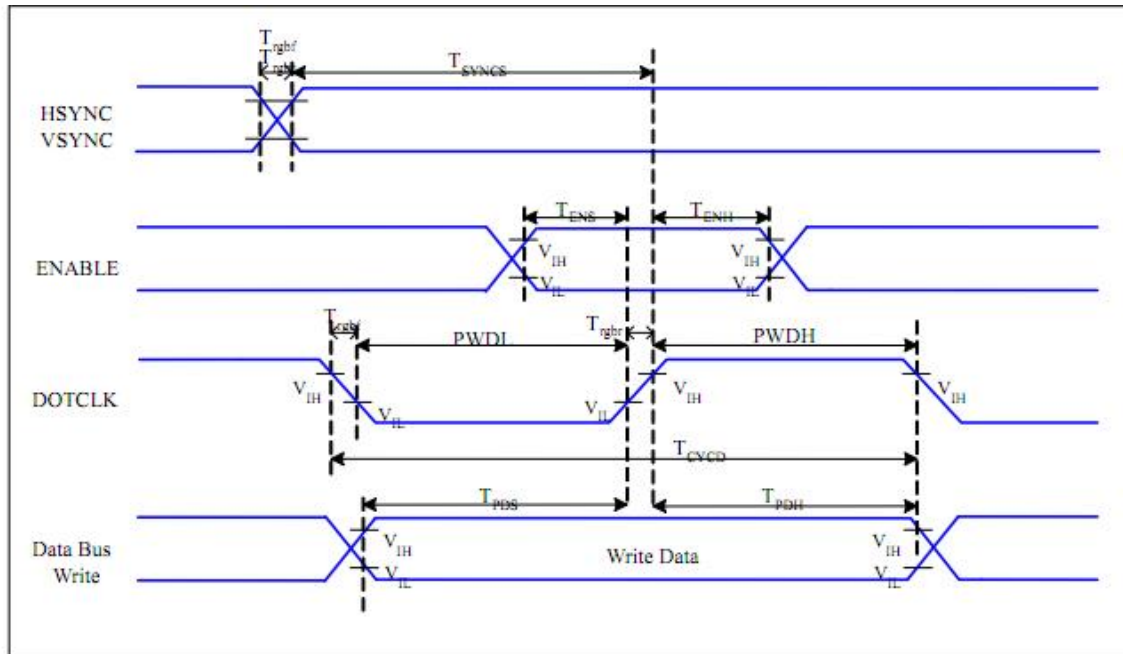


8.3. Serial Interface Characteristics (4-line serial)



Signal	Symbol	Parameter	MIN	MAX	Unit	Description
CSX	T_{CSS}	Chip select setup time (write)	15	-	ns	
	T_{CSH}	Chip select hold time (write)	15	-	ns	
	T_{CSS}	Chip select setup time (read)	60	-	ns	
	T_{SCC}	Chip select hold time (read)	65	-	ns	
	T_{CHW}	Chip select "H" pulse width	40	-	ns	
SCL	T_{SCYCW}	Serial clock cycle (Write)	16	-	ns	-write command & data ram
	T_{SHW}	SCL "H" pulse width (Write)	7	-	ns	
	T_{SLW}	SCL "L" pulse width (Write)	7	-	ns	
	T_{SCYCR}	Serial clock cycle (Read)	150	-	ns	-read command & data ram
	T_{SHR}	SCL "H" pulse width (Read)	60	-	ns	
	T_{SLR}	SCL "L" pulse width (Read)	60	-	ns	
D/CX	T_{DCS}	D/CX setup time	10	-	ns	
	T_{DCH}	D/CX hold time	10	-	ns	
SDA (DIN)	T_{SDS}	Data setup time	7	-	ns	
	T_{SDH}	Data hold time	7	-	ns	
DOUT	T_{ACC}	Access time	10	50	ns	For maximum CL=30pF
	T_{OH}	Output disable time	15	50	ns	For minimum CL=8pF

8.4. RGB Interface Characteristics:

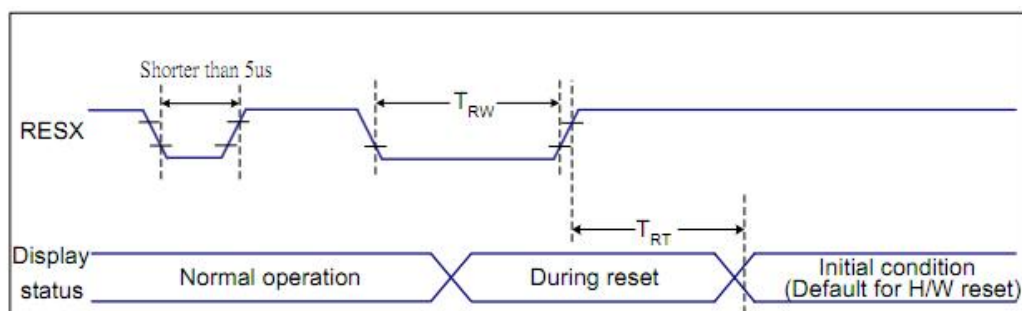


RGB Interface Timing Characteristics

Signal	Symbol	Parameter	MIN	MAX	Unit	Description
HSYNC, VSYNC	T_{SYNC}	VSYNC, HSYNC Setup Time	25	-	ns	
ENABLE	T_{ENS}	Enable Setup Time	25	-	ns	
	T_{ENH}	Enable Hold Time	25	-	ns	
DOTCLK	$PWDH$	DOTCLK High-level Pulse Width	25	-	ns	
	$PWDL$	DOTCLK Low-level Pulse Width	25	-	ns	
	T_{CYCD}	DOTCLK Cycle Time	55	-	ns	
	T_{rghr}, T_{rgbf}	DOTCLK Rise/Fall time	-	10	ns	
DB	T_{PDS}	PD Data Setup Time	25	-	ns	
	T_{PDH}	PD Data Hold Time	25	-	ns	

6 Bits RGB Interface Timing Characteristics

8.5. Reset Timing



Reset Timing

$VDDI=1.65$ to $3.3V$, $VDD=2.4$ to $3.3V$, $AGND=DGND=0V$, $T_a=25^{\circ}C$

Related Pins	Symbol	Parameter	MIN	MAX	Unit
RESX	TRW	Reset pulse duration	TBD	-	us
	TRT	Reset cancel	-	TBD (Note 1, 5)	ms
				TBD (Note 1, 6, 7)	ms

Reset Timing

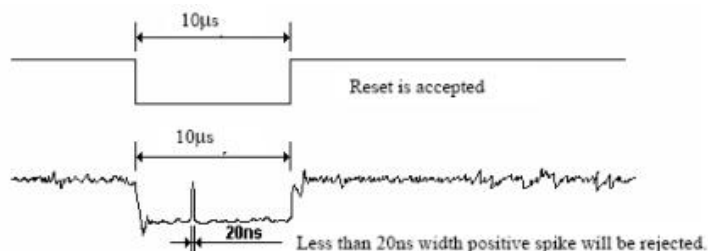
Notes:

- The reset cancel includes also required time for loading ID bytes, VCOM setting and other settings from NVM (or similar device) to registers. This loading is done every time when there is HW reset cancel time (t_{RT}) within 5 ms after a rising edge of RESX.
- Spike due to an electrostatic discharge on RESX line does not cause irregular system reset according to the table below:

RESX Pulse	Action
Shorter than 5us	Reset Rejected
Longer than 9us	Reset
Between 5us and 9us	Reset starts

- During the Resetting period, the display will be blanked (The display is entering blanking sequence, which maximum time is 120 ms, when Reset Starts in Sleep Out -mode. The display remains the blank state in Sleep In -mode.) and then return to Default condition for Hardware Reset.

- Spike Rejection also applies during a valid reset pulse as shown below:



- When Reset applied during Sleep In Mode.
- When Reset applied during Sleep Out Mode.
- It is necessary to wait 5msec after releasing RESX before sending commands. Also Sleep Out command cannot be sent for 120msec.



9. Quality Assurance

9.1. Purpose

This standard for Quality Assurance assures the quality of LCD module products supplied to customer.

9.2. Standard for Quality Test

9.2.1. Sampling Plan:

GB2828.1-2012

Single sampling, general inspection level II

9.2.2. Sampling Criteria:

Visual inspection: AQL 1.5%

Electrical functional: AQL 0.65%.

9.2.3. Reliability Test:

Detailed requirement refer to Reliability Test Specification.

9.3. Nonconforming Analysis & Disposition

9.3.1. Nonconforming analysis:

9.3.1.1. Customer should provide overall information of non-conforming sample for their complaints.

9.3.1.2. After receipt of detailed information from customer, the analysis of nonconforming parts usually should be finished in one week.

9.3.1.3. If cannot finish the analysis on time, customer will be notified with the progress status.

9.3.2. Disposition of nonconforming:

9.3.2.1. Non-conforming product over PPM level will be replaced.

9.3.2.2. The cause of non-conformance will be analyzed. Corrective action will be discussed and implemented.

9.4. Agreement Items

Shall negotiate with customer if the following situation occurs:

9.4.1. There is any discrepancy in standard of quality assurance.

9.4.2. Additional requirement to be added in product specification.

9.4.3. Any other special problem.

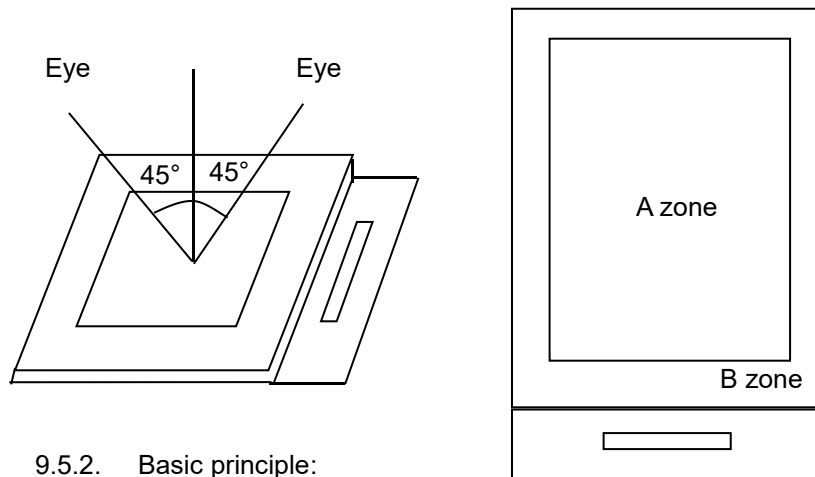
9.5. Standard of the Product Visual Inspection

9.5.1. Appearance inspection:

9.5.1.1. The inspection must be under illumination about 1000 – 1500 lx, and the distance of view must be at 30cm ± 2cm.

9.5.1.2. The viewing angle should be 45° from the vertical line without reflection light or follows customer's viewing angle specifications.

9.5.1.3. Definition of area: A Zone: Active Area, B Zone: Viewing Area,

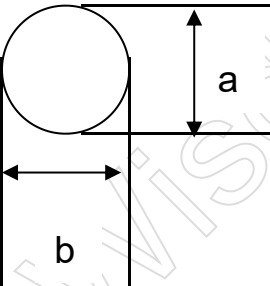


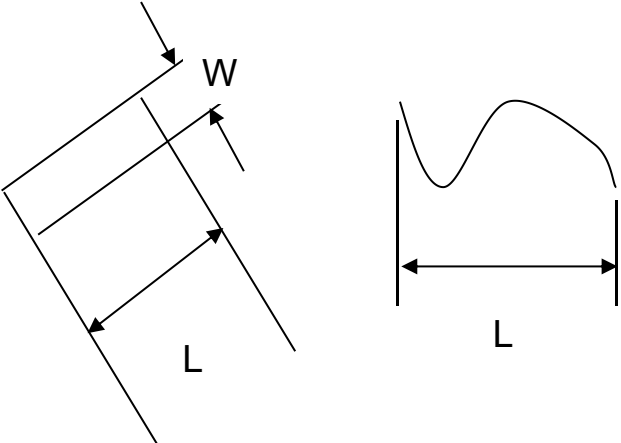
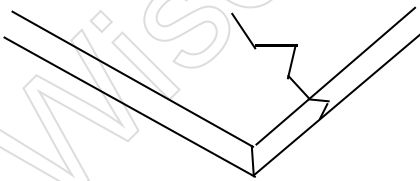
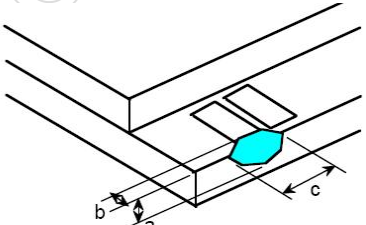
9.5.2. Basic principle:

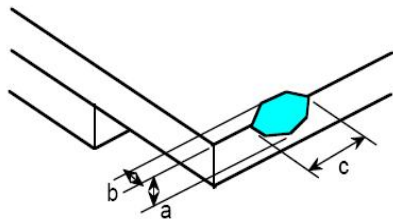
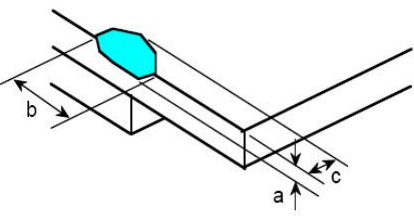
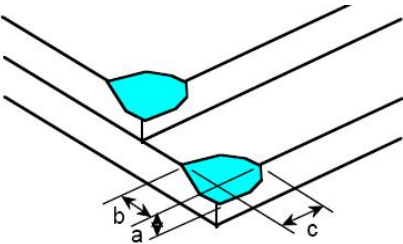
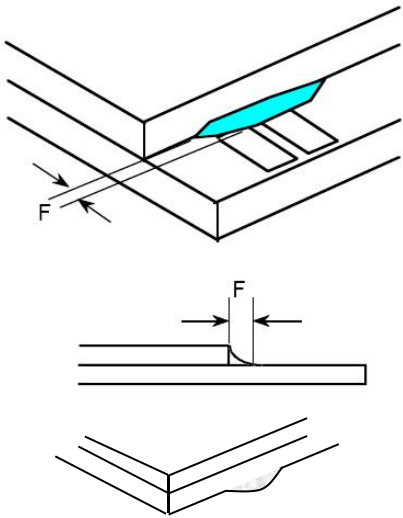
9.5.2.1. A set of sample to indicate the limit of acceptable quality level must be discussed by both us and customer when there is any dispute happened.

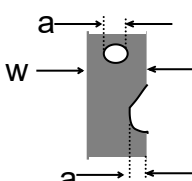
9.5.2.2. New item must be added on time when it is necessary.

9.6. Inspection Specification

No.	Item	Criteria (Unit: mm)																			
01	Black / White spot Foreign material (Round type) Pinholes Stain Particles inside cell. (Minor defect)	 $\varphi = (a + b) / 2$	<table><tr><th>Size</th><th>Area</th><th>Acc. Qty</th></tr><tr><td>$\varphi \leq 0.10$</td><td></td><td>Ignore</td></tr><tr><td>$0.10 < \varphi \leq 0.20$</td><td></td><td>$N \leq 2$</td></tr><tr><td>$0.20 < \varphi$</td><td></td><td>0</td></tr></table>			Size	Area	Acc. Qty	$\varphi \leq 0.10$		Ignore	$0.10 < \varphi \leq 0.20$		$N \leq 2$	$0.20 < \varphi$		0				
			Size	Area	Acc. Qty																
$\varphi \leq 0.10$		Ignore																			
$0.10 < \varphi \leq 0.20$		$N \leq 2$																			
$0.20 < \varphi$		0																			
		Distance between 2 defects should more than 5mm apart.																			
02	Electrical Defect (Minor defect)	<table><tr><td rowspan="2">Bright dot</td><td>Display Area</td><td>Total</td><td rowspan="4">Note 1</td></tr><tr><td>$N \leq 0$</td><td>$N \leq 0$</td></tr><tr><td>Dark dot</td><td>$N \leq 1$</td><td>$N \leq 1$</td></tr><tr><td>Total dot</td><td>$N \leq 1$</td><td>$N \leq 1$</td></tr><tr><td>Mura</td><td colspan="2">Not visible through 5% ND filters.</td><td>Note 2</td></tr></table>				Bright dot	Display Area	Total	Note 1	$N \leq 0$	$N \leq 0$	Dark dot	$N \leq 1$	$N \leq 1$	Total dot	$N \leq 1$	$N \leq 1$	Mura	Not visible through 5% ND filters.		Note 2
		Bright dot	Display Area	Total	Note 1																
			$N \leq 0$	$N \leq 0$																	
		Dark dot	$N \leq 1$	$N \leq 1$																	
		Total dot	$N \leq 1$	$N \leq 1$																	
		Mura	Not visible through 5% ND filters.		Note 2																
Remark:																					
1. Bright dot caused by scratch and foreign object accords to item 1.																					

03	Black and White line Scratch Foreign material (Line type) (Minor defect)	 <table border="1" data-bbox="585 687 1206 956"><thead><tr><th>Length</th><th>Width</th><th>Acc. Qty</th></tr></thead><tbody><tr><td>/</td><td>$W \leq 0.1$</td><td>Ignore</td></tr><tr><td>$L \leq 2$</td><td>$0.1 < W \leq 0.2$</td><td>2</td></tr><tr><td>$L > 2$</td><td>$0.2 < W$</td><td>0</td></tr><tr><td colspan="2">Total</td><td>2</td></tr></tbody></table> Distance between 2 defects should more than 3mm apart. Scratches not viewable through the back of the display are acceptable.	Length	Width	Acc. Qty	/	$W \leq 0.1$	Ignore	$L \leq 2$	$0.1 < W \leq 0.2$	2	$L > 2$	$0.2 < W$	0	Total		2
Length	Width	Acc. Qty															
/	$W \leq 0.1$	Ignore															
$L \leq 2$	$0.1 < W \leq 0.2$	2															
$L > 2$	$0.2 < W$	0															
Total		2															
04	Glass Crack (Minor defect)	 Crack is potential to enlarge, any type is not allowed.															
05	Glass Chipping Pad Area: (Minor defect)	 <table border="1" data-bbox="845 1538 1316 1713"><thead><tr><th>Length and Width</th><th>Acc. Qty</th></tr></thead><tbody><tr><td>$c > 3.0, b < 1.0$</td><td>1</td></tr><tr><td>$c < 3.0, b < 1.0$</td><td>2</td></tr><tr><td colspan="2">$a < \text{Glass Thickness}$</td></tr></tbody></table>	Length and Width	Acc. Qty	$c > 3.0, b < 1.0$	1	$c < 3.0, b < 1.0$	2	$a < \text{Glass Thickness}$								
Length and Width	Acc. Qty																
$c > 3.0, b < 1.0$	1																
$c < 3.0, b < 1.0$	2																
$a < \text{Glass Thickness}$																	

06	Glass Chipping Rear of Pad Area: (Minor defect) 	<table><tr><th>Length and Width</th><th>Acc. Qty</th></tr><tr><td>$c > 3.0, b < 1.0$</td><td>1</td></tr><tr><td>$c < 3.0, b < 1.0$</td><td>2</td></tr><tr><td>$c < 3.0, b < 0.5$</td><td>3</td></tr><tr><td colspan="2">$a < \text{Glass Thickness}$</td></tr></table>	Length and Width	Acc. Qty	$c > 3.0, b < 1.0$	1	$c < 3.0, b < 1.0$	2	$c < 3.0, b < 0.5$	3	$a < \text{Glass Thickness}$	
Length and Width	Acc. Qty											
$c > 3.0, b < 1.0$	1											
$c < 3.0, b < 1.0$	2											
$c < 3.0, b < 0.5$	3											
$a < \text{Glass Thickness}$												
07	Glass Chipping Except Pad Area: (Minor defect) 	<table><tr><th>Length and Width</th><th>Acc. Qty</th></tr><tr><td>$c > 3.0, b < 1.0$</td><td>1</td></tr><tr><td>$c < 3.0, b < 1.0$</td><td>2</td></tr><tr><td>$c < 3.0, b < 0.5$</td><td>3</td></tr><tr><td colspan="2">$a < \text{Glass Thickness}$</td></tr></table>	Length and Width	Acc. Qty	$c > 3.0, b < 1.0$	1	$c < 3.0, b < 1.0$	2	$c < 3.0, b < 0.5$	3	$a < \text{Glass Thickness}$	
Length and Width	Acc. Qty											
$c > 3.0, b < 1.0$	1											
$c < 3.0, b < 1.0$	2											
$c < 3.0, b < 0.5$	3											
$a < \text{Glass Thickness}$												
08	Glass Corner Chipping: (Minor defect) 	<table><tr><th>Length and Width</th><th>Acc. Qty</th></tr><tr><td>$c < 2.0, b < 2.0$</td><td>Ignore</td></tr><tr><td colspan="2">$a < \text{Glass Thickness}$</td></tr></table>	Length and Width	Acc. Qty	$c < 2.0, b < 2.0$	Ignore	$a < \text{Glass Thickness}$					
Length and Width	Acc. Qty											
$c < 2.0, b < 2.0$	Ignore											
$a < \text{Glass Thickness}$												
09	Glass Burr: (Minor defect) 	<table><tr><th>Length</th><th>Acc. Qty</th></tr><tr><td>$F < 1.0$</td><td>Ignore</td></tr></table> Glass burr don't affect assemble and module dimension.	Length	Acc. Qty	$F < 1.0$	Ignore						
Length	Acc. Qty											
$F < 1.0$	Ignore											

10	FPC Defect: (Minor defect) 	10.1 Dent, pinhole width $a < w/3$. (w: circuitry width.) 10.2 Open circuit is unacceptable. 10.3 No oxidation, contamination and distortion.								
11	Bubble on Polarizer (Minor defect)	<table><tr><th>Diameter</th><th>Acc. Qty</th></tr><tr><td>$\varphi \leq 0.20$</td><td>Ignore</td></tr><tr><td>$0.30 < \varphi \leq 0.40$</td><td>$N \leq 1$</td></tr><tr><td>$0.40 < \varphi$</td><td>$N = 0$</td></tr></table>	Diameter	Acc. Qty	$\varphi \leq 0.20$	Ignore	$0.30 < \varphi \leq 0.40$	$N \leq 1$	$0.40 < \varphi$	$N = 0$
Diameter	Acc. Qty									
$\varphi \leq 0.20$	Ignore									
$0.30 < \varphi \leq 0.40$	$N \leq 1$									
$0.40 < \varphi$	$N = 0$									
12	Dent on Polarizer (Minor defect)	<table><tr><th>Diameter</th><th>Acc. Qty</th></tr><tr><td>$\varphi \leq 0.15$</td><td>Ignore</td></tr><tr><td>$0.2 < \varphi \leq 0.30$</td><td>$N \leq 2$</td></tr><tr><td>$0.30 < \varphi$</td><td>None</td></tr></table>	Diameter	Acc. Qty	$\varphi \leq 0.15$	Ignore	$0.2 < \varphi \leq 0.30$	$N \leq 2$	$0.30 < \varphi$	None
Diameter	Acc. Qty									
$\varphi \leq 0.15$	Ignore									
$0.2 < \varphi \leq 0.30$	$N \leq 2$									
$0.30 < \varphi$	None									
13	Bezel	13.1 No rust, distortion on the Bezel. 13.2 No visible fingerprints, stains or other contamination.								
14	Touch Panel	D: Diameter W: width L: length 14.1 Spot: $D < 0.25$ is acceptable $0.25 \leq D \leq 0.4$ 2dots are acceptable and the distance between defects should more than 10 mm. $D > 0.4$ is unacceptable 14.2 Dent: $D > 0.40$ is unacceptable 14.3 Scratch: $W \leq 0.03$, $L \leq 10$ is acceptable, $0.03 < W \leq 0.10$, $L \leq 10$ is acceptable Distance between 2 defects should more than 10 mm. $W > 0.10$ is unacceptable.								
15	LCD Ripple	Touch the touch panel, cannot see the LCD ripple. Pen: R 0.8mm silicon rubber. Operation Force:120g								
16	PCB	16.1 No distortion or contamination on PCB terminals. 16.2 All components on PCB must same as documented on the BOM/component layout. 16.3 Follow IPC-A-600F.								

17	Soldering	Follow IPC-A-610C standard
18	Electrical Defect (Major defect)	The below defects must be rejected. 18.1 Missing vertical / horizontal segment, 18.2 Abnormal Display. 18.3 No function or no display. 18.4 Current exceeds product specifications. 18.5 LCD viewing angle defect. 18.6 No Backlight. 18.7 Dark Backlight. 18.8 Touch Panel no function.

Remark: LCD Panel Broken shall be rejected. Defect out of LCD viewing area is acceptable.

9.7. Classification of Defects

- 9.7.1. Visual defects (Except no / wrong label) are treated as minor defect and electrical defect is major.
- 9.7.2. Two minor defects are equal to one major in lot sampling inspection.

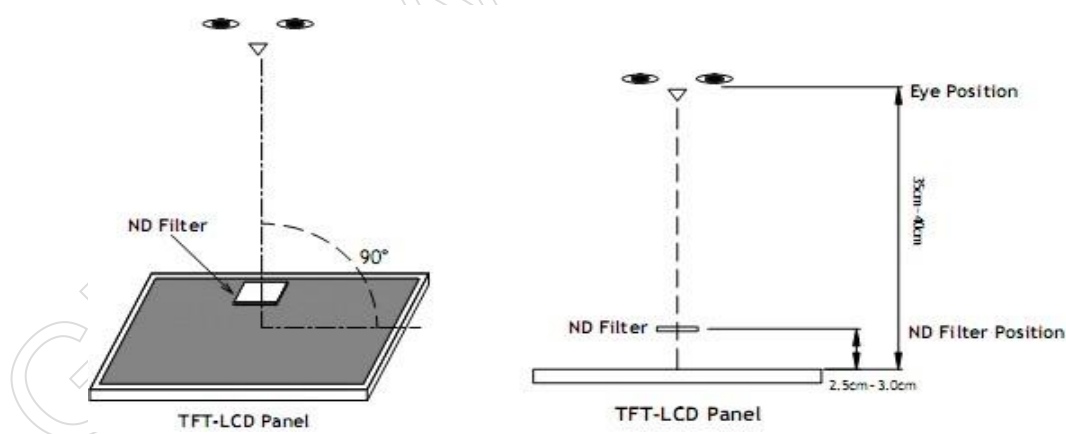
9.8. Identification/marketing criteria

Any unit with illegible / wrong /double or no marking/ label shall be rejected.

9.9. Packing

- 9.9.1. There should be no damage of the outside carton box, each packaging box should have one identical label.
- 9.9.2. Modules inside package box should have compliant mark.
- 9.9.3. All direct package materials shall offer ESD protection.

Note1: Bright dot is defined as the defective area of the dot is larger than 50% of one sub-pixel area.



Bright dot: The bright dot size defect at black display pattern. It can be recognized by 2% transparency of filter when the distance between eyes and panel is $350\text{mm} \pm 50\text{mm}$.

Dark dot: Cyan, Magenta or Yellow dot size defect at white display pattern. It can be recognized by 5% transparency of filter when the distance between eyes and panel is $350\text{mm} \pm 50\text{mm}$.

Note2: Mura on display which appears darker / brighter against background brightness on parts of display area.



10. Reliability Specification

No	Item	Condition	Quantity	Criteria
1	High Temperature Operating	Ts = +70℃, 96Hrs	2	IEC60068-2-2 GB2423.2-89
2	Low Temperature Operating	Ta = -20℃, 96Hrs	2	IEC60068-2-1 GB2423.1-89
3	High Humidity	Ta = 50℃, 90%RH, 96Hrs	2	IEC60068-2-3 GB/T2423.3-2006
4	High Temperature Storage	Ta = 80℃, 96Hrs	2	IEC60068-2-2 GB2423.2-89
5	Low Temperature Storage	Ta = -30℃, 96Hrs	2	IEC60068-2-1 GB2423.1-89
6	Thermal Cycling Test (non-operation)	-20℃, 60min~70℃, 60min, 20 cycles.	2	IEC60068-2-14 GB2423.22-87
7	Packing vibration	Frequency range:10Hz~50Hz Acceleration of gravity:5G X,Y,Z 30 min for each direction.	2	GB/T5170.14 -2009
8	Electrical Static Discharge	Air: ± 8KV 150pF/330 Ω 5 times Contact: ± 4KV 150pF/330 Ω 5 times	2	IEC61000-4-2 GB/T2423.5-1995
9	Drop Test (Packaged)	Height:80 cm,1 corner, 3 edges, 6 surfaces.	2	GB/T2423.8 -1995

Note1. After the reliability test, the product only guarantee function normally without any fatal defect (non-display, line defect, abnormal display). All the appearance inspection is judged before the reliability test (the product must have enough recovery time, at least 2 hours at room temperature);

Note2. Total current Consumption should be below double of initial value;

Note3. One product only can use to conduct one Item test;

Note4. Ts is the temperature of panel's surface. Ta is the ambient temperature of samples.

11. Precautions and Warranty

11.1. Safety

- 11.1.1. The liquid crystal in the LCD is poisonous. Do not put it in your mouth. If the liquid crystal touches your skin or clothes, wash it off immediately using soap and water.
- 11.1.2. Since the liquid crystal cells are made of glass, do not apply strong impact on them. Handle with care.

11.2. Handling

- 11.2.1. Reverse and use within ratings in order to keep performance and prevent damage.
- 11.2.2. Do not wipe the polarizer with dry cloth, as it might cause scratch. If the surface of the LCD needs to be cleaned, wipe it swiftly with cotton or other soft cloth soaked with petroleum IPA, do not use other chemicals.

11.3. Storage

- 11.3.1. Do not store the LCD module beyond the specified temperature ranges.
- 11.3.2. Strong light exposure causes degradation of polarizer and color filter

11.4. Metal Pin (Apply to Products with Metal Pins)

11.4.1. Pins of LCD and Backlight

13.4.1.1 Solder tip can touch and press on the tip of Pin LEAD during the soldering

13.4.1.2 Recommended Soldering Conditions

Solder Type: Sn96.3~94-Ag3.3~4.3-Cu0.4~1.1

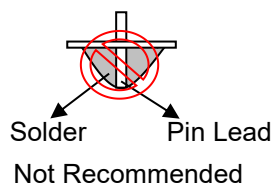
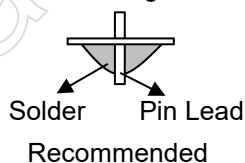
Maximum Solder Temperature: 370°C

Maximum Solder Time: 3s at the maximum temperature

Recommended Soldering Temp: 350±20°C

Typical Soldering Time: ≤3s

13.4.1.3 Solder Wetting



11.4.2. Pins of EL

13.4.2.1 Solder tip can touch and press on the tip of EL leads during soldering.

13.4.2.2 No Solder Paste on the soldering pad on the motherboard is recommended.

13.4.2.3 Recommended Soldering Conditions

Solder type: Nippon Alimit Leadfree SR-34, size 0.5mm

Recommended Solder Temperature: 270~290°C

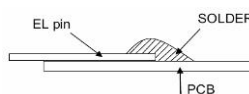
Typical Soldering Time: ≤2s

Minimum solder distance from EL lamp (body):2.0mm

13.4.2.4 No horizontal press on the EL leads during soldering.

13.4.2.5 180° bend EL leads three times is not allowed.

13.4.2.6 Solder Wetting

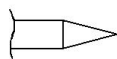


Recommended

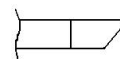


Not Recommended

13.4.2.7 The type of the solder iron:

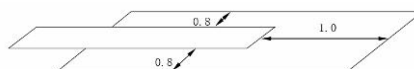


Recommended



Not Recommended

13.4.2.8 Solder Pad



11.5. Operation

- 11.5.1. Do not drive LCD with DC voltage
- 11.5.2. Response time will increase below lower temperature
- 11.5.3. Display may change color with different temperature
- 11.5.4. Mechanical disturbance during operation, such as pressing on the display area, may cause the segments to appear "fractured".
- 11.5.5. Do not connect or disconnect the LCM to or from the system when power is on.
- 11.5.6. Never use the LCM under abnormal condition of high temperature and high humidity.
- 11.5.7. Module has high frequency circuits. Sufficient suppression to the electromagnetic interface shall be done by system manufacturers. Grounding and shielding methods may be important to minimize the interference.
- 11.5.8. Do not display the fixed pattern for long time (we suggest the time not longer than one hour) because it may develop image sticking due to the TFT structure.

11.6. Static Electricity

- 11.6.1. CMOS LSIs are equipped in this unit, so care must be taken to avoid the electro-static charge, by ground human body, etc.
- 11.6.2. The normal static prevention measures should be observed for work clothes and benches.
- 11.6.3. The module should be kept into anti-static bags or other containers resistant to static for storage.

11.7. Limited Warranty

- 11.7.1. Our warranty liability is limited to repair and/or replacement. We will not be responsible for any consequential loss.
- 11.7.2. If possible, we suggest customer to use up all modules in six months. If the module storage time over twelve months, we suggest that recheck it before the module be used.
- 11.7.3. After the product shipped, any product quality issues must be feedback within three months, otherwise, we will not be responsible for the subsequent or consequential



events.

12. Packaging

TBD

Global Wisdom Display